INCH-POUND

A-A-52212
October 23, 1996
SUPERSEDING
MIL-G-21196B
20 February 1968
(see 6.5)

COMMERCIAL ITEM DESCRIPTION

GAGE, CYLINDER

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers one type of cylinder gage.

1.2 Intended use. The gage is used to measure engine cylinder bores for straightness, out of roundness, and taper diameters.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The cylinder gage shall be similar to figure 1, and shall have a range of 2 1/2 to 9 inches. The cylinder gage shall have a dial indicator attached to a mounting block. The mounting block shall be attached to a slide base to permit alignment of the gage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5210

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The gage shall have a locking device which is fastened to the body of the mounting block. The locking device shall lock the adjustable contact point on the indicator dial to a threaded hole in the mounting block. A toggle joint shall be provided with a toggle joint handle that can be threaded into the toggle joint. The handle shall be 8 1/2 inches $\pm$ 1 inch in length and 5/16 inch $\pm$ 1/8 inch in diameter. The handle shall have a knurled outer surface with one end threaded for attachment to the block. The other end shall have a knurled cap screw. Extension rods shall be provided that slide into an opening in the lower section of the cylinder block. The rods shall be held in the gage spindle by a method that provides for easy insertion and removal of the rods. The rods shall be of sufficient length to measure the 0.001 inch increments within the 2 1/2 inch to 9 inch range of the cylinder gage.

FIGURE 1. Cylinder gage with dial mounted on block with locking device

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2.2 Dial indicator. The dial indicator utilized with this gage shall conform to type B, style III of A-A-2348 (see 6.2).

2.3 Hardness. All parts of the cylinder gage including the extension rods (except dial indicator parts) shall have a minimum hardness of Rockwell "C" 55 when tested as specified in 4.2.1.

2.4 Finish. All exterior finished surfaces of the cylinder gage, except the gaging surfaces, shall have a surface roughness not greater than 63 micro inches arithmetical average (A.A.) when measured as specified in 4.2.2. The gaging surfaces shall have a surface roughness not greater than 8 micro inches (A.A.) when measured as specified in 4.2.2.

2.5 Marking for identification. The gage shall be marked for identification in accordance with the manufacturer's standard practice. The marking shall be permanent and legible.

2.6 Workmanship. The finished gage shall be complete, clean and free of fabrication marks, scratches, sharp edges, deformities, breaks, sharp corners and slivers. The gage shall have no visible evidence of rust, dirt or other extraneous materials. There shall be no defects affecting serviceability, durability, operation, performance or appearance.

2.7 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of ASTM E 380, and all other requirements of this CID are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3. REGULATORY REQUIREMENTS

3 .1 Contractor recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, shall conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product that has been sold in the commercial marketplace for over one year, or is the same product that has successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years.

b. The company must have sold 200 items meeting the requirements of this CID, in the commercial or government marketplace over the past 2 years.

4.2 Tests.

4.2.1 Hardness test. All parts of the cylinder gage including the extension rods (excluding dial indicator parts) shall be tested for hardness in accordance with ASTM E 18. Failure of any part of the cylinder gage to meet the hardness requirement specified in 2.3 shall constitute a test failure of the cylinder gage.

4.2.2 Surface roughness. All exterior finished surfaces and the gaging surfaces of the cylinder gage shall be examined for surface roughness using commercially recognized instruments for surface roughness measurement. Surfaces not meeting the requirements specified in 2.4 shall constitute failure of the examination.

5. PACKAGING

5.1 Packaging. Preservation, packing and marking requirements shall be as specified in the contract or purchase order (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred option permitted herein and include the following information in the procurement document (if applicable).

a. Title, number and date of this CID
b. Packaging requirements (see 5.1)

6.2 Sources for documents

6.2.1 Sources for Government publications.

A-A-2348 - Indicator, Dial, Accessories and Test Sets

(Copies of this CID may be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

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6.2.2 Sources for non-government association documents.

ASTM STANDARDS
E18 - Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E380 - Use of the International System of Units (SI) (The Modernized Metric System)

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428).

6.3 National Stock Number (NSN). The NSN assigned to the gage, cylinder covered by this CID is 5210-00-494-1774.

6.4 Sources of supply. Manufacturers whose products are known to meet the requirements of this CID are listed below; however, competition is not limited to these companies:

Products Engineering Corporation
Post Office Box 45009
9430 Bellanca Avenue
Los Angeles, CA 90045

L. S. Starrett
121 Crescent Street
Athol, MA 01331

6.5 Deleted items. The following gage that appeared in MIL-G-21196B has been deleted in this CID since there are no user requirements for the item:

Type II - Dial mounted on slide base with setting gage

MILITARY INTERESTS:

Custodians
Army - GL
Navy - YD1
Air Force - 99

Review Activities

Army - AT, AV, MI
Navy - MC, NP, SH
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:

ARMY - GL

Project 5210-0295

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